* NOTE:
1. VALUES MARKED * ARE ESTIMATES — VERIFY WITH DATASHEET / ON SITE.
2. DG SET / CHANGEOVER: NONE CONSIDERED. CONFIRM ON SITE; IF A DG EXISTS, ADD CHANGEOVER + INTERLOCK (SOLAR OFF ON DG).

## CABLE DETAILS

| CABLE SIZE | CURRENT RATING | CABLE CONNECTION | CABLE TYPE |
|---|---|---|---|
| 1.1 KV, 1R X 3.5C X 95 SQ MM | 216 A ** | INVERTER TO ACDB | AL. XLPE INSULATED ARMOURED CABLE |
| 1.1 KV, 1R X 3.5C X 95 SQ MM | 216 A ** | ACDB TO CLIENT LT PANEL | AL. XLPE INSULATED ARMOURED CABLE |
| 1R X 1C X 4 SQ MM | 55 A ** | MODULE TO DCDB / INVERTER | XLPO UV Cu. CABLE FOR +VE & –VE |
| 1R X 1C X 16 SQ MM | — | PV STRUCTURE TO EP-1 | Cu EARTH CABLE |
| 1R X 1C X 25 SQ MM | — | INVERTER / ACDB TO EP-2 | Cu EARTH CABLE |
| 1R X 1C X 50 SQ MM | — | LA TO EP-3 | Cu EARTH CABLE |

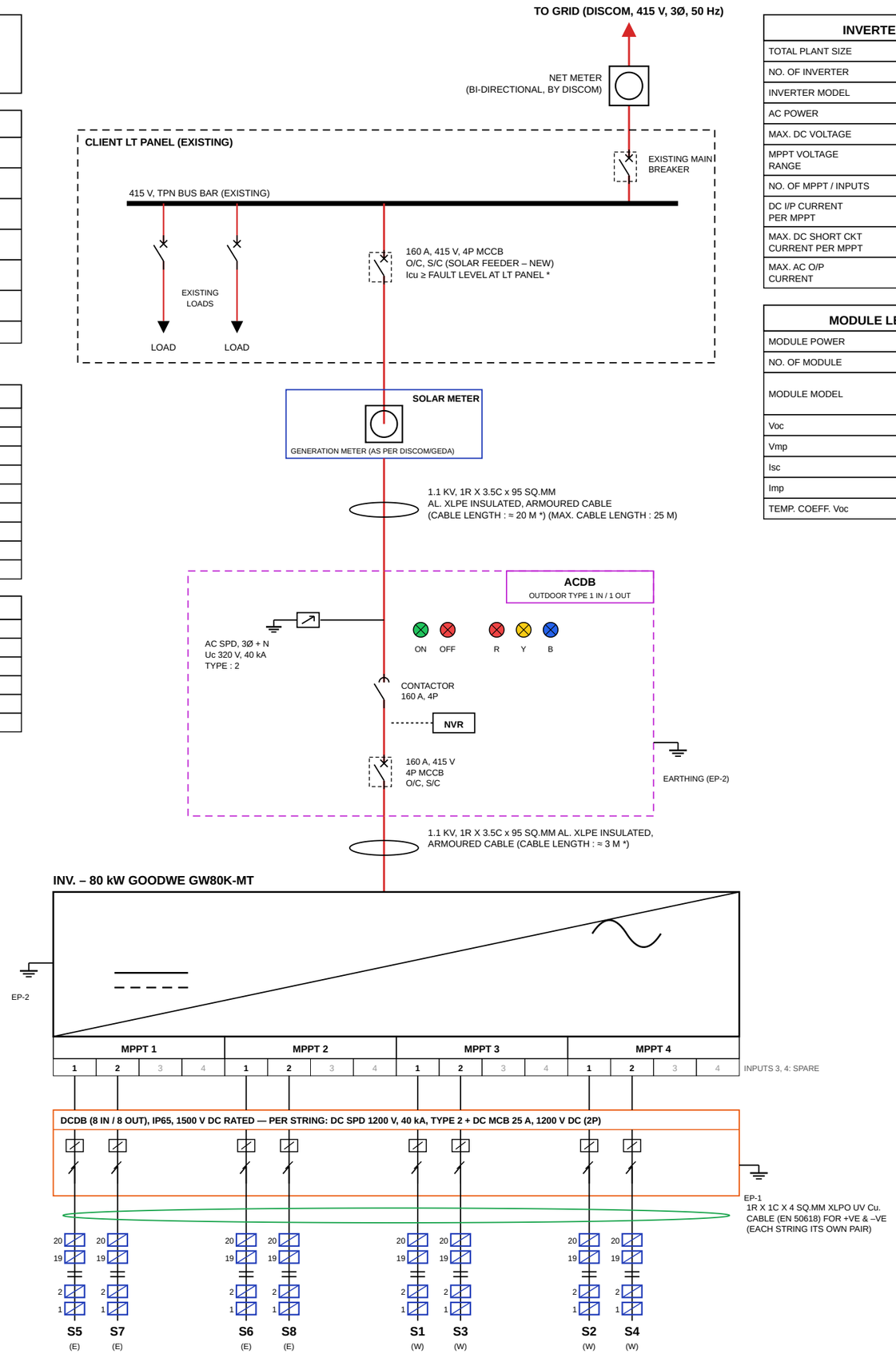
** CATALOGUE RATING IN AIR. DESIGN CURRENT: AC 1.25 x 127 A = 159 A; DC 1.25 x 15.96 A = 20 A PER STRING. CHECK WITH CABLE MAKER'S TABLE.

## STRING / MPPT SCHEDULE

| MPPT | INPUT | STRING | MODULES | SLOPE |
|---|---|---|---|---|
| 1 | 1 | S5 | 20 | EAST |
| 1 | 2 | S7 | 20 | EAST |
| 2 | 1 | S6 | 20 | EAST |
| 2 | 2 | S8 | 20 | EAST |
| 3 | 1 | S1 | 20 | WEST |
| 3 | 2 | S3 | 20 | WEST |
| 4 | 1 | S2 | 20 | WEST |
| 4 | 2 | S4 | 20 | WEST |

## DESIGN CHECK

| | |
|---|---|
| Voc @ 5 °C (20 x 52.3 V) | 1,046 V < 1,100 V ✓ |
| Vmp @ 70 °C / STC / 5 °C | 723 / 832 / 880 V (200–1000 V) ✓ |
| CURRENT PER MPPT (Imp / Isc) | 30.1 A ≤ 44 A / 31.9 A ≤ 55 A ✓ |
| DC VOLTAGE DROP (4 SQ MM) | 0.52 % – 1.03 % * |
| AC VOLTAGE DROP (95 SQ MM AL) | ≈ 0.45 % * |
| STRING FUSE | NOT REQUIRED (≤ 2 STRINGS / MPPT) |




## INVERTER LEGEND

| | |
|---|---|
| TOTAL PLANT SIZE | 100.00 kWp |
| NO. OF INVERTER | 1 |
| INVERTER MODEL | GW80K-MT |
| AC POWER | 80 kW |
| MAX. DC VOLTAGE | 1100 V |
| MPPT VOLTAGE RANGE | 200 – 1000 V |
| NO. OF MPPT / INPUTS | 4 / 16 |
| DC I/P CURRENT PER MPPT | 44 A |
| MAX. DC SHORT CKT CURRENT PER MPPT | 55 A |
| MAX. AC O/P CURRENT | 127 A * |

## MODULE LEGEND (STC)

| | |
|---|---|
| MODULE POWER | 625 Wp |
| NO. OF MODULE | 160 |
| MODULE MODEL | ADANI AB-G12R-132-625 (N-TYPE TOPCon) |
| Voc | 49.8 V * |
| Vmp | 41.6 V * |
| Isc | 15.96 A * |
| Imp | 15.03 A * |
| TEMP. COEFF. Voc | –0.25 %/°C * |

## SYMBOLS NOMENCLATURE

| | |
|---|---|
| | PV MODULE |
| | STRING INVERTER |
| | EARTHING |
| | SURGE PROTECTION DEVICE (SPD) |
| | MINIATURE CIRCUIT BREAKER (MCB) |
| | MOULDED CASE CIRCUIT BREAKER (MCCB) |
| | SOLAR ENERGY GENERATION METER |
| | NET METER (DISCOM) |
| | R, Y, B LIVE INDICATING LAMPS |
| | CIRCUIT BREAKER ON |
| | CIRCUIT BREAKER OFF |
| | CONTACTOR |
| NVR | NO VOLT RELAY |
| | CABLE |

## SYSTEM DESCRIPTION

| | |
|---|---|
| SITE LOCATION | — (LAT, LONG TO BE ADDED) |
| DC CAPACITY | 100.00 kWp |
| TOTAL PLANT CAPACITY (AC) | 80 kW |
| DC / AC RATIO | 1.25 |
| MODULE CAPACITY | 625 Wp (ADANI N-TOPCon) |
| MODULE DIMENSION (L X W X H) | 2382 X 1134 X 30 mm * |
| TOTAL PV MODULES | 160 Nos. |
| INVERTER CAPACITY | 80 kW (GOODWE GW80K-MT) 1 No. |
| AZIMUTH / TILT | EAST-WEST (90° / 270°), ROOF SLOPE |

INVERTER PROTECTION :- 80 kW (GOODWE GW80K-MT)
1. DC REVERSE POLARITY PROTECTION
2. AC SHORT CIRCUIT PROTECTION
3. INSULATION RESISTANCE DETECTION
4. RESIDUAL-CURRENT MONITORING UNIT (RCMU)
5. ANTI-ISLANDING PROTECTION
6. AC / DC SURGE PROTECTION TYPE II
7. DC SWITCH (INTEGRATED)
8. STRING CURRENT MONITORING
9. AC OVER / UNDER VOLTAGE & FREQUENCY PROTECTION
10. ANTI-PID / AFCI : OPTIONAL
11. SAFETY STANDARDS : AS PER GOODWE DATASHEET (IEC 62109-1/-2, IEC 61727, IEC 62116)

CUSTOMER SIGN AND STAMP:

| REV. | DATE | DESCRIPTION | DRAWN BY | CHECKED BY | APPROVED BY |
|---|---|---|---|---|---|
| R0 | 27-10-26 | FIRST ISSUE | | | |

| | |
|---|---|
| CLIENT: | **JAIN GRANITE & MARBLE** |
| EPC CONTRACTOR: | **NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara |
| DESIGN BY: | **NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara |

DRAWING NO.: 01

DRAWING NAME: **SINGLE LINE DIAGRAM**

PROJECT TITLE: 100.00 kWp (DC)_ROOFTOP SOLAR PLANT

RELEASED FOR: ☐ PRELIMINARY ☐ TENDER ☐ REVISION ☐ FABRICATION ☐ CONSTRUCTION ■ APPROVAL

SCALE: **NTS**

| DESIGN BY | DRAWN BY | CHECKED BY | APPROVED BY | PAPER SIZE | SHEET NO. |
|---|---|---|---|---|---|
| 27-10-26 | 27-10-26 | 27-10-26 | 27-10-26 | **A3** | **01 OF 01** |